TM-001: Requirements Traceability Matrix

Project: Project Atlas (Master Flightline Clock)
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Purpose

This document maps functional requirements to software modules and provides a framework for future verification and validation.

Current Traceability

ID	Requirement	Primary Module(s)	Status	Verification
URS-001	Display accurate local system time.	engine.py	Implemented	Clock updates continuously from system time.
URS-002	Maintain heat schedule.	database.py, models.py	Implemented	SQLite persistence.
URS-003	Determine operational phase automatically.	engine.py	Implemented	P0-P4 state logic.
URS-004	Adaptive display layouts.	led_simulator.py	Implemented	64x64 and 128x64 rendering.
URS-005	Operator schedule management.	windows.py	Implemented	Add/Edit/Delete heats.
URS-006	Persist settings.	database.py, config.py	Implemented	Settings table.
URS-007	Brightness control.	windows.py, led_simulator.py	Implemented	Live brightness updates.
URS-008	Hardware profile selection.	windows.py	Planned	Settings drop-down (LCD / 64x64 / 128x64).
URS-009	Automatic appliance startup.	launcher.py (planned)	Planned	LaunchAgent / Startup configuration.

URS-010	Exit to desktop.	windows.py (planned)	Planned	Administrative command.

Future Verification

Each requirement will later be linked to formal test procedures (TP-xxx), integration tests and acceptance tests to provide full GAMP-style traceability.

Architect Notes

- No business logic should depend on display hardware.
- The timing engine remains the single source of truth.
- Hardware-specific behaviour belongs only in the rendering/output layer.
- Deployment targets include macOS and Windows.